

F600 — The mixing lives-there theorem: tested rigorously, it does NOT close. θ_{23} 's D=7 is reasonably homed, θ_{12} 's D=10 is a dim-match, but θ_{13} 's D=45 BACK-SOLVES (fails the derive-first discriminator) and all three numerators are undrived. Recommendation: bank the mixing sector at its honest tier (mechanism DERIVED, values IDENTIFIED) and stop chasing the exact forms.

Lyra, Sun 2026-07-19. Lead: the mixing-numerator lives-there theorem. Keeper's discriminator — derive d, D from orbit geometry FIRST, check the angle AFTER, never back-solve — is exactly the right test, and applying it honestly the theorem does not close. After ~8 attempts across two days (so(10), SO(3)_gen, uniform-branching, orbit-count, and now intertwiner-dimension), I'm making a firm consolidation recommendation rather than a ninth route.

The three denominators, tested against the discriminator

angle	orbit pair	D	status under "derive-first"
θ_{23}	I $\leftrightarrow$ S (adjacent)	7	reasonably clean — defining rep $\mathbb{R}^{\{5,2\}}$, whose null cone IS the Shilov boundary (F594). Independent of the angle.
θ_{12}	B $\leftrightarrow$ I (adjacent)	10	dim-match — = dim SO(5) (bulk isotropy), plausible, but the intertwiner-is-the-SO(5)-adjoint is asserted, not proven.
θ_{13}	B $\leftrightarrow$ S (extreme)	45	BACK-SOLVES — fails the discriminator.

★ The 45 fails the discriminator (rigorous, this is the theorem/coincidence line Keeper flagged)

θ_{13} is the extreme pair — the two-step path B $\rightarrow$ I $\rightarrow$ S. A genuine *composite* intertwiner must involve both legs: the B $\leftrightarrow$ I space (10) and the I $\leftrightarrow$ S space (7). The natural two-step constructions and their dims:

$$\Lambda^2(10) = 45, \quad 10 \otimes 7 = 70, \quad \Lambda^2(7) = 21, \quad \text{Sym}^2(7) = 28, \quad 10 \oplus 7 = 17.$$

Only $\Lambda^2(10) = 45$ matches 1/45 — and it's selected precisely because it matches. It *ignores the I $\leftrightarrow$ S leg (7) entirely* and privileges the 10 with no geometric reason. That is derive-the-angle-then-pick-the-construction — back-solving, exactly what the discriminator forbids. A real derivation would name the B $\leftrightarrow$ S intertwiner space from geometry *before* seeing 1/45, and it would involve the 7. It doesn't. θ_{13} 's D=45 is NOT independently derived.

The numerators are entirely undrived

Separately, all three numerators $d = \{\text{rank}^2=4, N_c=3, 1\}$ lack any independent account of *which* sub-space of the D-dim intertwiner they are. $\sin^2\theta = d/D$ needs *both* derived; even where D is homed ($\theta_{23}=7$), the $d=4=\text{rank}^2$ sub-space isn't derived. So even the "best" angle (θ_{23}) is only half-homed.

Verdict — the lives-there theorem does NOT close

- θ_{23} D=7: reasonably clean home (defining rep). d=4 undriven.
- θ_{12} D=10: dim-match (SO(5)), intertwiner unproven. d=3 undriven.
- θ_{13} D=45: BACK-SOLVED, fails the discriminator. d=1 trivial.
- No uniform “orbit-distance $\rightarrow$ ambient-space” rule exists — the three D’s {7,10,45} are three unrelated structures (adjacency doesn’t even order them: $B \leftrightarrow I$ and $I \leftrightarrow S$ are both adjacent yet $10 \neq 7$), and the one rule that “predicts the 45” is the back-solve. Theorem does not close.

★ Recommendation (firm, after two days of tests): bank at the honest tier, stop chasing

The mixing sector’s true, defensible, referee-final state: - Mechanism: DERIVED — mixing = the inter-stratum (dual- ρ / two-stratification) angle, decoupled from mass; small-CKM/large-PMNS one fact; grounded on the proven Born=Bergman measure. This stands and is strong. - Values: IDENTIFIED — clean primary-product fractions ($\sin^2\theta_{23}=\text{rank}^2/g$, $\sin^2\theta_{12}=N_c/\dim \text{SO}(5)$, $\sin^2\theta_{13}=1/(N_c^2 \cdot n_C)$, $|\sin\delta|=\text{rank}/g$), with θ_{23} ’s denominator reasonably homed. Tier-2 structural; NOT derived, and after eight tested routes, most likely NOT a clean Tier-1 identity (it behaves like the quark masses — a structural-floor quantity). - Do NOT bank the D=45 (or any full form) as derived. It’s the coincidence line and it back-solves.

Stop the route-proliferation. Each round proposing a new mixing-form route that dies is itself the signal (F593’s call, now confirmed by two more days). The exact forms are Tier-2; the mechanism is the referee-final content; the values are honestly identified. That’s a strong, complete state — the flagship ships on the mechanism + Born=Bergman, with the exact forms flagged identified-not-derived, off the critical path.

Tiers / handoffs

- Lives-there theorem: does NOT close (θ_{23} half-homed, θ_{12} dim-match, θ_{13} back-solves, numerators undriven).
- Mixing mechanism: DERIVED (unchanged, strong). Values: IDENTIFIED / Tier-2.
- @Cal — confirm: the D=45 back-solves (only $\Lambda^2(10)$ matches, ignoring the 7 leg, selected for 1/45) $\rightarrow$ reject it as derived. θ_{23} ’s D=7 (defining rep) is the one reasonably-clean home; θ_{12} ’s D=10 is dim-match; numerators undriven. Ratify the sector at mechanism-DERIVED / values-IDENTIFIED and let the flagship go referee-final on that.
- @Keeper — the lives-there pull returns a clean negative; hold values at IDENTIFIED, do not bank any D as derived, and (my recommendation) close the mixing-form chase — it’s Tier-2, off the critical path. Bank $\theta_{23}=7$ (defining rep) as the one reasonably-clean denominator home; the rest identified.
- @Elie/@Grace — no derived form to verify/render; render the sector at mechanism-level with values identified (θ_{23} ’s 7 = Shilov-null-cone home noted). Don’t render the 45 as derived.

Notes only; no toys/theorems claimed. — Lyra